Pragati

Property Appreciation Intelligence for India

Founder's plan: market landscape, product blueprint, modules,
business model, Day 1 / 90 / 360 roadmap, VC fundraising, BRD and PRD

Working draft v0.1 · 23 October 2026

"Pragati" is a placeholder name. Figures are planning hypotheses.

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Pragati (working name): Property Appreciation Intelligence for India

Give it an area or a building and it estimates how much the property is likely to appreciate over the next 1–5 years, based on the infrastructure and development projects announced around it. Every forecast is explained, linked to government and trusted sources, and back-tested.

Founder's summary

- **Does this already exist in India?** Partly. Portals (99acres, MagicBricks, Housing.com, Square Yards, NoBroker) show backward-looking asking-price trends. Research firms (Liases Foras, PropEquity, Anarock, Knight Frank, JLL) publish city-level reports. Registration-data players (Square Yards, Zapkey, PropStack) sell transaction history. **Nobody offers forward-looking, building-level forecasts that are explicitly linked to upcoming developments, with published accuracy.** That is the wedge.
- **Positioning:** neutral (no selling leads to developers), explainable, and source-linked.
- **Business model:** B2C freemium (paid reports + Pro subscription) to build brand and data, then B2B and API (developers, banks/HFCs/NBFCs, funds) for venture scale.
- **Roadmap:** Day 1 = one city, hand-curated catalysts, map, analyst-assisted reports. Day 90 = real models, explainable forecasts, payments, B2B design partners. Day 360 = 6–8 cities, completion-probability model, B2B console and API, land/plot forecasts.
- **Funding path:** grants/angels, then pre-seed (~₹1–4 Cr) on one-city traction, then seed ($1.5–4M) on multi-city revenue and proven accuracy. Pitch it as *data infrastructure / lender risk*, not only "proptech".

Documents

Doc	What's inside
01 — Market landscape	Indian & global competitors, the gap, why now, risks
02 — Product blueprint	Screens, all modules, forecasting approach, architecture, Day 1 / 90 / 360, future use cases
03 — Business model & fundraising	Segments, pricing, unit economics, GTM, moats, VC strategy and pitch deck outline
BRD	Business objectives, scope, stakeholders, business requirements, KPIs, risks
PRD	Personas, user stories with acceptance criteria, model requirements, data model, NFRs, release plan

All pricing, targets and market numbers are planning hypotheses to validate, not verified facts. Check competitor details and fund theses before using them externally.

01 — Market Landscape: Who Already Does This in India?

> Short answer: **many players do pieces of this, but nobody in India does "development-linked, explainable, back-tested appreciation forecasts" for a specific building or micro-market as their core product.** That gap is the opportunity.

Status, pricing and features of the companies below change often. Verify each one (product page, recent funding news, a test account) before quoting them in a pitch deck.

1. Competitive map

Category	Examples	What they do well	What they don't do
Listing portals with price trends	99acres, MagicBricks (PropIndex), Housing.com / PropTiger, NoBroker, Square Yards	Huge traffic, asking-price trends by locality, some "locality insights"	Prices are mostly *asking* prices, backward-looking, not tied to upcoming infrastructure, no forward forecast with confidence
Registration / transaction data	Square Yards (IGR-based transaction data), Zapkey, PropStack, CRE Matrix (commercial)	Actual registered transaction values, owner history	Historical only; sold mainly as data to B2B; little forecasting
Research & analytics firms	Liases Foras, PropEquity, Anarock Research, Knight Frank, JLL, CBRE, Colliers	Credible city/micro-market reports, supply/absorption data, developer-side relationships	Quarterly PDFs, city-level granularity, expensive, not self-serve, not per-building
Official indices	RBI House Price Index, NHB RESIDEX	Trusted, transaction-based	Very coarse (city level), lagging, no forecasts
Buyer advisory / curated platforms	Propsoch and similar "buyer-side advisor" startups	Data-led advice to home buyers	Mostly advisory/brokerage revenue; forecasting not the core productised IP
Fractional / investment platforms	Strata, hBits, Alt DRX-style platforms	Investor UX, yield framing	Commercial-asset focused, not a forecasting engine
Bank/NBFC valuation	Empanelled valuers, in-house risk teams	Regulated collateral valuation	Manual, slow, no forward risk/appreciation view

Global reference points (useful for the pitch)

- **Zillow Zestimate / Redfin Estimate (US)** — automated valuation models (AVMs) on current value. They prove the consumer demand for "what's my home worth", but they are about *today's* value, not development-driven *future* value.

- **HouseCanary, CoreLogic, Cotality (US)** — forecasting and AVMs sold to lenders and investors. The B2B playbook in this repo copies this model.
- **Local Logic (Canada)** — location scores built from geospatial data. Close to our "catalyst map" concept.

2. The gap we exploit

1. **Forward-looking, not backward-looking.** Everyone shows what prices *did*. Nobody credibly shows what a specific metro line, expressway, airport, IT park or master-plan rezoning *will do* to a specific building.
2. **Explainability.** "₹X/sq ft by 2029 because the Purple Line extension station is 650 m away (62% probability of opening by 2028), the ring road is in tender stage, and 4,200 new units are coming in the same micro-market."
3. **Evidence and trust.** Every catalyst is linked to its government source document, and the model publishes its own back-test accuracy. Portals have a conflict of interest because they earn from developers and brokers; we don't sell leads (see business model).
4. **Granularity.** Building/project level and 500 m grid level, not "Pune West".

3. Why now

- **RERA (2016)** made project data (approvals, timelines, carpet area, complaints) public per state.
- **State IGR portals** (Maharashtra IGR, Karnataka Kaveri, Telangana/AP registration, etc.) expose registration and circle-rate data.
- **Infrastructure push**: PM Gati Shakti National Master Plan, metro expansion in 20+ cities, Bharatmala expressways, new airports (Navi Mumbai, Noida/Jewar), and Smart Cities give a large, visible pipeline of catalysts.
- **Cheap geospatial and LLM tooling** makes it possible to read tender PDFs, master plans and gazette notifications at scale. Five years ago this would have needed a large team of analysts.

4. Key risks highlighted by the landscape

Risk	Mitigation
Portals copy the feature	Moat = curated catalyst database + back-test track record + neutrality; portals can't credibly claim neutrality
Data access (IGR scraping, ToS)	Prefer official bulk data / RTI / licensed partners (e.g. license from a transaction-data vendor at start); legal review per state
Forecast wrong → reputational/legal risk	Ranges not point estimates, published accuracy, strong disclaimers, "information not advice" positioning
Indian real estate is opaque (cash component, under-reporting)	Blend registration values, listing prices, and developer price sheets; model the gap explicitly

02 — Product Blueprint

Working name: **Pragati** ("progress"). It's a placeholder; check trademark and domain availability before committing to it.

One-liner: *Type any building, project or locality in India and see how much it is likely to appreciate over the next 1–5 years, which upcoming developments drive that, and how confident we are.*

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1. What the platform looks like

1.1 Home / search

- A single search bar: "Search a building, project, RERA ID, locality or drop a pin".
- City switcher, plus a "Hot catalysts this month" feed (e.g. "Metro Line 3 Phase 2 tender awarded").

1.2 Property / area report page (the core screen)

```
┌──────────────────────────────────────────────────────────────────┐
│  Lodha Amara, Kolshet Road, Thane | RERA P51700012345 | Residential  │
├──────────────────────────────────┬───────────────────────────────┤
│  TODAY'S ESTIMATED VALUE         │  5-YEAR APPRECIATION OUTLOOK  │
│  Rs 14,200 / sq ft               │  Base case +38%  (Rs 19,600 / sq ft)  │
│  (range 13,600 - 14,900)         │  Bear +18%        Bull +55%   │
│  Based on 46 registrations       │  Confidence: Medium (3 of 5)  │
├──────────────────────────────────┴───────────────────────────────┤
│  [Chart] Price history 2015-today, forecast fan chart to 2031    │
├──────────────────────────────────────────────────────────────────┤
│  WHAT DRIVES THIS FORECAST                 impact   probability  │
│  [METRO] Line 4, Kasarvadavali stn, 900 m  +9.5%    78% by 2027  │
│  [ROAD]  Thane Coastal Road, 2.1 km        +4.0%    45% by 2029  │
│  [JOBS]  IT park (MIDC notification), 3 km +2.5%    60%          │
│  [RISK]  Supply overhang: 6,800 new units  -3.0%    -            │
│  [TREND] City-wide market trend            +22%     -            │
│  Each row links to its source (tender, gazette, DPR, RERA filing)  │
├──────────────────────────────────────────────────────────────────┤
│  MAP: catalysts, timelines, isochrones, flood/zoning overlays    │
├──────────────────────────────────────────────────────────────────┤
│  RISKS: project litigation, RERA complaints, flood zone, CRZ     │
│  COMPARABLES: 5 nearby projects, their price and forecast        │
│  ACCURACY: "3-yr forecasts in Thane within +/-7% in 71% of cases"  │
└──────────────────────────────────────────────────────────────────┘
  [Download PDF report]  [Set alert]  [Compare]  [Add to portfolio]
```

1.3 Catalyst explorer (map)

Every known infrastructure and policy project on a map, filterable by type (metro, road, airport, industrial corridor, rezoning, SEZ, IT park, hospital or university) and by lifecycle stage. Clicking a catalyst shows its timeline, budget, source documents, delay history, and the properties it affects.

1.4 Compare & portfolio

- Compare up to 4 properties or localities side by side.

- Portfolio: users add properties they own and track estimated value plus forecast changes.
- Alerts: "A catalyst affecting your property changed stage" (e.g. tender awarded, land acquisition stalled).

1.5 B2B console (developers, lenders, funds)

- Bulk upload of addresses or loan books → valuation, forecast and risk flags.
- Land/site-selection heatmaps: "Where will appreciation be highest in the next 5 years within 30 km of the CBD?"
- API access and white-label PDF reports.

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2. Modules

A. Data acquisition layer

#	Module	Sources	Notes
A1	Infrastructure catalyst ingestion	Metro rail corporations, NHAI / MoRTH, PM Gati Shakti, state PWDs, airport authorities, Smart Cities Mission, CPPP / state e-procurement tenders, Union & state budgets, cabinet decisions, gazette notifications	Track the **lifecycle**: Proposed → DPR → Approved → Funded → Land acquisition → Tendered → Awarded → Under construction → Commissioned
A2	Master plan & zoning	Development plans / regional plans (e.g. DDA MPD-2041, MMRDA, BDA/BMRDA, HMDA, PMRDA), TP schemes, FSI/TDR policies	Georeference plan PDFs; rezoning is a big catalyst
A3	Transactions & prices	State IGR registration data, circle rates / guidance values, licensed transaction data vendors, listing portals (asking prices), developer price sheets	Normalise to ₹/sq ft carpet area
A4	Project supply	State RERA portals (project, promoter, timeline, units, complaints), developer launches	Supply pipeline and project risk
A5	Macro & demand	RBI (rates, HPI), NHB RESIDEX, census and population projections, employment hubs (SEZ, IT parks), office absorption reports	Market baseline
A6	Geospatial	OpenStreetMap, Bhuvan (ISRO), satellite imagery (construction progress), flood maps, CRZ, eco-sensitive zones	Distance, travel time, risk overlays
A7	News & signals	Trusted news outlets, press releases (PIB), court orders (land-acquisition litigation)	Detects delays and stage changes

B. Intelligence layer

#	Module	What it does
B1	Document AI pipeline	LLM + OCR extraction from tenders, DPRs, gazettes and plan PDFs into structured catalyst records (type, location geometry, budget, dates, stage). Human-in-the-loop review queue before anything goes live

#	Module	What it does
B2	Entity resolution	Map "Hiranandani Estate", RERA IDs, survey numbers and addresses to one canonical property / building / micro-market ID
B3	Geo engine	Geocoding, distance and travel-time (isochrone) to each catalyst, 500 m H3 grid
B4	Current-value model (AVM)	Hedonic + gradient-boosted model on registrations and listings; gives today's value with a range
B5	Catalyst impact model	Learns historical uplift from past projects (event-study / difference-in-differences: e.g. what happened to prices within 1 km of Delhi/Bengaluru/Mumbai metro stations from announcement to opening, compared with control areas). Outputs uplift curves by catalyst type × distance × city tier × lifecycle stage
B6	Completion-probability model	Probability and expected delay of each catalyst, based on agency track record, stage, funding, land-acquisition status and litigation
B7	Forecast engine	Combines market baseline + Σ (catalyst uplift × completion probability × time profile × distance decay) − supply overhang − risk penalties → P10/P50/P90 trajectory, 1–5 years
B8	Explainability	Breaks the forecast into per-driver contributions (SHAP-style) and links every driver to its source
B9	Back-testing & accuracy	Re-runs historical forecasts ("what would we have said in 2018?") and publishes error metrics by city

Forecast formula (simplified):

```
Forecast(t) = Value_today
            × (1 + MarketBaseline(city, segment, t))
            × Π_i (1 + Uplift_i(type, distance, stage, t) × P_i(complete by t))
            × (1 − SupplyOverhangPenalty(t))
            × (1 − RiskPenalty)
```

Run as Monte Carlo over catalyst completion dates to get the fan chart.

C. Product layer

#	Module
C1	Web app (search, report page, catalyst map, compare, portfolio)
C2	Report generator (PDF, shareable link, white-label)
C3	Alerts & notifications (email, WhatsApp)
C4	Accounts, subscriptions, payments (Razorpay / UPI, GST invoicing)
C5	B2B console + bulk upload
C6	Public API (keys, metering, rate limits)
C7	Admin / analyst console (catalyst review queue, data QA, overrides with audit log)

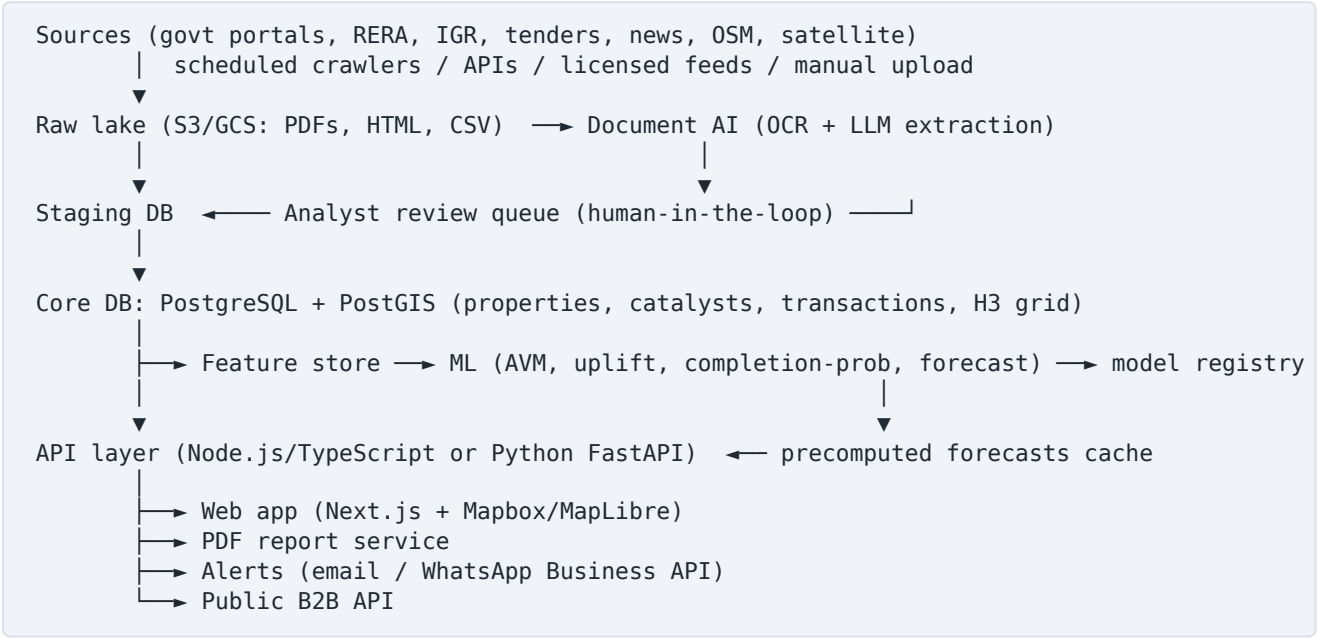
D. Trust, legal & ops layer

#	Module
D1	Source provenance: every data point stores URL, document hash, fetch date

#	Module
D2	Disclaimers and "information, not investment advice" framing; terms of use
D3	DPDP Act 2023 compliance (consent, data minimisation for user portfolios)
D4	Data licensing register (what we may redistribute, per source)
D5	Model governance: versioning, drift monitoring, accuracy dashboards

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3. Reference architecture



Suggested stack: Next.js + TypeScript, PostgreSQL/PostGIS, Python for ML (LightGBM, PyMC/Prophet for baselines), Airflow/Dagster for pipelines, an LLM for document extraction, MapLibre for maps, Razorpay for payments, hosted on AWS Mumbai (ap-south-1) for data residency.

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4. Day 1 vs Day 90 vs Day 360

Day 1 (MVP, "concierge + map")

Goal: prove that people want this and will pay for it.

- **One city, 10–15 micro-markets.** Pick a city with heavy, visible infrastructure activity and good data access (Pune, Hyderabad, Bengaluru or MMR; MMR has the richest IGR data).
- Catalyst database of the **top 50–100 projects in that city**, entered by hand from government sources, each with a stage and source links.
- Catalyst map + locality page: price history (registrations + listings), nearby catalysts, a simple rule-based "appreciation outlook" (low/medium/high plus an indicative % range).
- Paid **PDF report** (₹499–₹999), assembled semi-manually by an analyst within 24 hours.
- Waitlist, WhatsApp channel, weekly "catalyst update" newsletter.
- Team: founder(s) + 1 full-stack + 1 data/analyst.

Day 90 ("real model, first revenue")

- Building/project-level pages for all RERA-registered projects in the city.
- AVM v1 + catalyst uplift model v1 trained on historical metro and road projects in 3–4 cities; P10/P50/P90 forecasts.
- Explainability panel (driver contributions) and source links on every driver.
- Back-test v1 published for the launch city.
- Self-serve payments: single report + monthly Pro subscription.
- Alerts on catalyst stage changes.
- Document-AI pipeline that pulls in new tenders and notifications automatically, with analyst review.
- 2–3 **B2B design partners** (a mid-size developer for land selection, a broker network for white-label reports, one HFC/NBFC for collateral risk).
- Targets (hypotheses to validate): 20k MAU, 500 paid reports, 3 signed pilots.

Day 360 ("multi-city platform")

- 6–8 cities (MMR, Pune, Bengaluru, Hyderabad, Delhi-NCR, Chennai, Ahmedabad, plus one tier-2 growth city such as Nagpur or Lucknow).
- Completion-probability model per agency, Monte Carlo fan charts, supply-overhang model.
- Portfolio tracking, compare, NRI-focused flows (USD/AED view).
- B2B console with bulk upload; public API with metered billing.
- Land and plot forecasting (not only apartments), including TP schemes and rezoning.
- Accuracy leaderboard by city, refreshed quarterly.
- Targets (hypotheses): 250k MAU, ₹40–60 lakh MRR, 10+ paying B2B accounts, at least one bank/HFC in production.

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5. Future use cases (Year 2+)

1. **Lender risk products**: forward-looking LTV, collateral-decline early warning, LAP portfolio stress tests for banks, HFCs and NBFCs.
2. **Developer land intelligence**: "where to buy land next", demand forecasting per configuration and price point.
3. **Government / urban planning analytics**: value-capture financing estimates (betterment levy, TOD premium FSI) for metro corporations and municipal bodies.
4. **Insurance**: flood/climate risk combined with value for home insurers.
5. **Fractional and REIT/InvIT investors**: asset-level forecasts and portfolio analytics.
6. **NRI investor product**: remote due diligence, title and RERA checks, appreciation outlook.
7. **Rental yield forecasting** and rent-vs-buy calculators.
8. **Commercial, warehousing and data-centre land** forecasting along industrial corridors (DMIC, freight corridors).
9. **Agricultural → NA land conversion** opportunity scoring on city peripheries.
10. **Embedded data**: forecast widgets inside listing portals, banks' home-loan apps and wealth-management apps.
11. **"Infra delay index"**: a public index of government project delays (good for PR and policy influence).
12. **Marketplace (careful)**: verified, neutral transaction services (legal, title search, home loans) — only where it doesn't compromise forecast neutrality.

03 — Business Model & VC Fundraising Plan

All numbers below are **planning hypotheses**, not market facts. Validate them with pilots before putting them in a deck.

1. Core principle: neutrality is the product

Indian portals make most of their money from developers and brokers, so their "insights" are not trusted. **We do not sell buyer leads to developers or brokers in Years 1–2.** Revenue comes from people who need the *truth*: buyers, investors, lenders and the developers' own land teams.

2. Customer segments & revenue streams

Segment	Pain	Product	Pricing (hypothesis)
Home buyers (end users)	"Am I overpaying? Will this area grow?"	Free locality score; paid detailed report	Free tier; ₹499–₹1,999 per report
Retail investors & NRIs	Picking the next growth corridor remotely	Pro subscription: unlimited reports, alerts, portfolio, compare	₹999/month or ₹7,999/year; NRI tier ~$15/month
Brokers / channel partners / wealth advisors	Need credible, branded material to advise clients	White-label reports, client portfolios	₹5k–₹25k/month per seat bundle
Developers (land & strategy teams)	Site selection, pricing, launch timing	B2B console: heatmaps, micro-market forecasts, supply pipeline	₹10–50 lakh/year per developer
Banks, HFCs, NBFCs	Collateral valuation and forward risk on loan books	Bulk/API valuation + forecast + risk flags	Per-call (₹20–₹200) + annual platform fee ₹25 lakh–₹1 Cr
Funds, AIFs, REITs, family offices	Deal sourcing, underwriting	Data API, custom research	₹15–75 lakh/year
Government / metro corporations (later)	Value-capture financing, TOD planning	Custom analytics	Project-based

Revenue mix target: Year 1 ≈ 60% B2C/prosumer + 40% B2B pilots; Year 3 ≈ 25% B2C + 75% B2B/API. B2C builds the brand, data feedback loop and distribution. B2B builds revenue scale and the venture-scale story.

3. Unit economics (targets to validate)

Metric	Target
B2C CAC (content/SEO-led)	< ₹300 per paying user
Report gross margin	> 85% once automated

Metric	Target
Pro subscriber LTV	₹6,000–₹10,000 (12–15 month retention)
B2B ACV	₹15–50 lakh
B2B sales cycle	3–6 months (developers), 6–12 months (banks)
Data cost	Largest COGS line: licensed transaction data plus analyst review

4. Go-to-market

1. **SEO and content moat**: one auto-generated, high-quality page per project, locality and catalyst (100k+ pages). Searches like "Pune metro line 3 impact on Hinjewadi property prices" have intent and little good content.
2. **"Catalyst tracker" newsletter and WhatsApp channel**: weekly government infrastructure updates per city. Cheap audience building.
3. **YouTube / Instagram explainers**: "This ₹12,000 cr expressway will change these 5 localities."
4. **Broker and wealth-advisor partnerships**: they distribute our reports to their clients (white-label).
5. **NRI communities** (Gulf, US, UK): remote due diligence is a strong need.
6. **B2B founder-led sales**: start with 2–3 mid-size developers and one HFC as design partners. Offer free pilots in exchange for case studies and data.

5. Moats (what investors will probe)

1. **Curated catalyst graph**: years of structured, lifecycle-tracked government project data with source provenance. Hard and boring to replicate.
2. **Historical impact dataset**: learned uplift curves from hundreds of past Indian infrastructure projects.
3. **Published track record**: back-tested and live forecast accuracy. Trust compounds over time.
4. **Neutrality brand**: portals can't copy this without hurting their own revenue model.
5. **Distribution**: SEO page network + B2B integrations (API embedded in lender workflows is sticky).

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6. How to get VC funding

6.1 Stage plan

Stage	When	What you need to show	Typical size (India, indicative)
Bootstrapped / grants	Day 0–90	MVP in one city, waitlist, first paid reports	Startup India Seed Fund Scheme (via incubators), state startup grants, angel cheques
Pre-seed	~Day 90–180	Working forecasts in 1 city, 1–2 B2B pilots, early revenue, strong team	₹1–4 Cr (~$150k–$500k)
Seed	~Month 12–18	4–8 cities, ₹25–50 lakh MRR or strong B2B pipeline, back-test accuracy, retention data	$1.5–4M
Series A	~Month 24–36	₹5–10 Cr+ ARR, B2B/API majority, multiple lenders in production	$8–15M

6.2 Who to approach

- **Angel networks & syndicates**: Indian Angel Network, LetsVenture, Venture Catalysts, Inflection Point Ventures, and real-estate operators who angel-invest (they bring developer access).
- **Pre-seed / seed funds**: early-stage funds with a B2B SaaS, fintech or data thesis, e.g. Blume Ventures, 100X.VC, Better Capital, Antler India, Titan Capital, Java Capital, Kae Capital, Chiratae, Prime Venture Partners.
- **Larger seed / Series A**: Accel, Elevation, Peak XV (Surge), Lightspeed, Matrix/Z47, Nexus. Lead with a fintech/lender-risk angle; the "proptech" label alone has been a harder sell in India.
- **Strategics**: proptech arms of developers, housing finance companies, and global proptech funds (e.g. Fifth Wall, MetaProp) once you have traction.
- **Accelerators**: Y Combinator, Antler, Surge, NASSCOM 10,000 Startups, and IIT/IIM incubators (also a route to SISFS grants).

Check each fund's current thesis and portfolio for conflicts (e.g. they back a listing portal) before pitching.

6.3 Narrative for the deck (12 slides)

1. **Problem**: Indians put most of their wealth into real estate and buy with almost no forward-looking data. ₹1–2 Cr decisions are made on broker opinions.
2. **Insight**: Government infrastructure is the biggest driver of appreciation, and it is *public* but unstructured (PDFs, tenders, gazettes).
3. **Solution**: Explainable, source-linked, back-tested appreciation forecasts at building level.
4. **Demo**: report page and catalyst map.
5. **Why now**: RERA, IGR digitisation, Gati Shakti infra pipeline, LLM document extraction.
6. **Market**: Indian residential sales are worth several lakh crore rupees a year (use a cited source such as Anarock/Knight Frank annual figures); plus outstanding housing and LAP loan books for lenders. Build TAM → SAM → SOM bottom-up from paying segments.
7. **Business model**: B2C freemium → B2B/API.
8. **Traction**: users, paid reports, pilots, LOIs, accuracy metrics.
9. **Competition**: the map from the Market Landscape section, positioning on two axes: *backward vs forward-looking* × *conflicted vs neutral*.
10. **Moat**: catalyst graph, impact dataset, track record.
11. **Team**: real-estate domain + data/ML + GTM. A credible domain co-founder (ex-research firm, ex-developer land team, ex-lender risk) matters a lot here.
12. **Ask & use of funds**: e.g. 18–24 month runway; ~45% engineering/data, ~25% data acquisition & analysts, ~20% GTM, ~10% legal/ops.

6.4 Metrics VCs will ask for

- Forecast accuracy (MAPE / hit-rate within ±X%) by city, back-tested and live.
- Free → paid conversion, report repeat rate, Pro retention.
- B2B pilot → paid conversion, ACV, sales cycle length.
- Data coverage: % of RERA projects covered, number of catalysts tracked, freshness (days from government announcement to our database).
- CAC by channel, SEO traffic growth.

6.5 Red flags to pre-empt

Investor concern	Answer
"Forecasts can be wrong, so liability"	Ranges + confidence, published accuracy, information-not-advice terms, legal review
"Portals will copy you"	Neutrality + track record + catalyst graph; and portals could become API customers
"Data access is fragile"	Multiple sources per field, licensed feeds, RTI, government MoUs, own historical archive
"Consumer real estate is infrequent"	Recurring B2B/API revenue; portfolio tracking & alerts for owners keep B2C engaged
"Is the market venture-scale?"	Lender/insurer risk analytics and API (US analogues: HouseCanary, CoreLogic)

Business Requirements Document (BRD)

Field	Value
Product	Pragati (working name): development-linked property appreciation intelligence for India
Version	0.1 (initial draft)
Status	Draft for founder review
Related docs	PRD, Market landscape, Blueprint, Business model

1. Executive summary

Indian households keep most of their wealth in real estate, yet purchase decisions rely on broker opinion and backward-looking asking-price trends. Public infrastructure (metro, expressways, airports, industrial corridors, rezoning) is the biggest driver of future appreciation, and information about it is public but scattered across thousands of unstructured government documents. Pragati turns that information into explainable, source-linked, back-tested appreciation forecasts for any building, project or locality.

2. Business problem

1. Buyers and investors cannot quantify how upcoming developments affect a specific property.
2. Existing portals show asking prices and are financially tied to developers and brokers, so their numbers aren't trusted as neutral.
3. Research firms publish city-level PDFs that are expensive, infrequent and not per-building.
4. Lenders value collateral manually and have no forward view of collateral risk.

3. Business objectives

#	Objective	Measure	Target (Month 12)
BO1	Become the trusted neutral source for forward-looking property data in launch cities	MAU, NPS, branded search volume	250k MAU, NPS > 40
BO2	Prove willingness to pay	Paid reports + Pro subscribers	10k paid reports cumulative; 2,000 Pro subscribers
BO3	Establish the B2B revenue line	Paying B2B accounts, ARR	10 accounts, ₹3 Cr+ ARR run-rate
BO4	Prove forecast credibility	Published back-test accuracy	≥ 65% of 3-year back-tests within ±10%
BO5	Raise institutional capital	Seed round	Seed closed by Month 15–18

4. Scope

In scope (Year 1)

- Residential apartments and projects (RERA-registered) in 6–8 cities; plots on city peripheries from Day 360.
- Infrastructure, zoning and supply catalysts from government and trusted sources.
- Web app (responsive), PDF reports, alerts, B2C subscriptions, B2B console and API (beta).

Out of scope (Year 1)

- Brokerage, selling leads to developers, or transacting property.
- Regulated investment advice or portfolio management.
- Native mobile apps (a responsive web app/PWA comes first).
- Commercial office/retail forecasting (Year 2).

5. Stakeholders

Stakeholder	Interest
Founders / leadership	Vision, fundraising, prioritisation
Data & ML team	Forecast quality, pipelines
Research analysts	Catalyst curation, QA
Engineering	Platform delivery
GTM (growth, B2B sales)	Acquisition, revenue
Legal / compliance advisor	Data licensing, disclaimers, DPDP, RERA implications
Customers: buyers, investors, NRIs, brokers, developers, lenders, funds	See segments in business model doc
Data sources: government agencies, data vendors	Access, licensing, attribution

6. Business requirements

ID	Requirement	Priority
BR-01	Provide a current-value estimate with a range for any covered property or locality	Must
BR-02	Provide a 1–5 year appreciation forecast with bear/base/bull scenarios and a confidence level	Must
BR-03	Explain every forecast by listing the drivers (catalysts, market trend, supply, risks) and their contribution	Must
BR-04	Link every catalyst to its original source document(s), with fetch date	Must

ID	Requirement	Priority
BR-05	Track the lifecycle stage of each catalyst and update it as new information arrives	Must
BR-06	Publish forecast accuracy (back-tested and live) per city	Must
BR-07	Monetise through paid reports and subscriptions (UPI/cards, GST-compliant invoices)	Must
BR-08	Offer B2B bulk valuation/forecast and an API with usage metering	Should (Day 90 pilot, Day 360 GA)
BR-09	Notify users when catalysts affecting saved properties change	Should
BR-10	Support white-label reports for brokers and advisors	Should
BR-11	Maintain neutrality: no paid placement or developer influence on forecasts	Must (policy)
BR-12	Comply with DPDP Act 2023, source terms of use, and consumer-protection norms	Must

7. Success metrics (KPIs)

- **North star:** weekly users who view a full forecast report.
- Acquisition: organic traffic, SEO pages indexed, newsletter subscribers.
- Monetisation: free → paid conversion (target 2–4%), ARPU, B2B ARR.
- Trust: accuracy metrics, report-dispute rate, NPS.
- Data: catalysts tracked, % RERA projects covered, median "announcement → in database" lag (target < 72 hours).

8. Assumptions

1. Enough historical transaction and price data can be licensed or collected to train uplift models (at least 3 cities × 8+ years).
2. Users will pay for a neutral, explainable report before a ₹50 lakh+ decision.
3. Lenders and developers will run pilots with an early-stage vendor if the data is demonstrably better.
4. Government sources remain publicly accessible.

9. Constraints & dependencies

- Data-licensing terms per state IGR and portal; possible restrictions on redistribution.
- Analyst capacity for human-in-the-loop review in the early phase.
- LLM/OCR accuracy on scanned regional-language documents (Marathi, Kannada, Telugu, Hindi, etc.).
- Mapping and geocoding quality in Indian addresses.

10. Risks

Risk	Impact	Likelihood	Mitigation
Inaccurate forecasts damage trust	High	Medium	Ranges, confidence, published accuracy, conservative launch cities
Legal claims from users who relied on forecasts	High	Low–Medium	Clear ToS, "information not advice", no guaranteed returns language, legal review
Data access blocked or restricted	High	Medium	Multiple sources, licensing, own archive, RTI
Incumbent portals launch a similar feature	Medium	High	Neutrality, depth, track record, B2B focus
Slow B2B sales cycles	Medium	High	Start with mid-size developers and NBFCs; land-and-expand
Defamation risk on project risk flags	Medium	Low	Only factual, sourced flags (RERA complaints, court orders)

11. High-level timeline

Phase	Milestone
Day 0–30	Launch city chosen, top 100 catalysts curated, MVP locality pages, waitlist
Day 30–90	AVM v1, uplift model v1, paid reports, Pro plan, 2–3 B2B design partners, first back-test
Day 90–180	3 cities, document-AI pipeline, alerts, pre-seed raise
Day 180–360	6–8 cities, B2B console + API, completion-probability model, seed-round preparation

12. Approval

Role	Name	Sign-off
Founder / CEO		
Co-founder / CTO		
Advisor (real estate)		
Advisor (legal)		

Product Requirements Document (PRD)

Field	Value
Product	Pragati (working name)
Version	0.1 (MVP → Day 360)
Owner	Founder / Product
Related	BRD, Blueprint

1. Problem statement

"I'm considering buying in [project/locality]. What is it worth today, how much will it appreciate over the next few years because of what's being built around it, and how sure can I be?" No product in India answers this credibly at building level.

2. Goals & non-goals

Goals

- G1: Answer "value today + 1–5 year outlook + why" for any covered property in under 10 seconds.
- G2: Make every number traceable to data and sources.
- G3: Convert intent into revenue (reports, Pro, B2B).

Non-goals (v1)

- Brokerage, listings marketplace, home-loan origination.
- Point-precise "guaranteed" predictions.
- Native mobile apps.

3. Personas

Persona	Description	Top job-to-be-done
Ananya, first-time buyer (32, Pune, IT)	Choosing between 3 projects, ₹80 L–1.2 Cr budget	"Which option holds and grows its value best?"
Rajesh, investor (45, business owner)	Owns 3 properties, wants the next growth corridor	"Where should I buy land/flats for 5-year appreciation?"
Farah, NRI (38, Dubai)	Buying remotely in Hyderabad	"Can I trust this project and this area from abroad?"
Vikram, channel partner	Advises 200+ clients per year	"Give me credible, branded reports to share"
Meera, land-acquisition head at a developer	Evaluates 20 land parcels per quarter	"Which parcel will see the highest demand/price growth?"

Persona	Description	Top job-to-be-done
Arjun, risk head at an HFC	40k home loans on the book	"Which collateral is at risk of value decline? What's a fair forward LTV?"

4. User stories & requirements

Priority: **P0** = Day 1/MVP, **P1** = Day 90, **P2** = Day 360.

4.1 Search & discovery

ID	User story	Acceptance criteria	P
S-1	As a user, I can search by project name, RERA ID, locality, or address	Autocomplete < 300 ms; fuzzy matching; top-5 results include the right entity for ≥ 95% of test queries	P0
S-2	I can drop a pin on the map and get a locality/grid report	Pin → H3 cell → report in < 3 s	P0
S-3	I see a "hot catalysts" feed for my city	Shows the 10 latest stage changes with dates and sources	P1

4.2 Property / locality report

ID	User story	Acceptance criteria	P
R-1	I see the estimated current price/sq ft with a range	Shows P50 and P10–P90 range; number of comparable transactions; data recency	P0 (locality), P1 (building)
R-2	I see a 1/3/5-year appreciation outlook with bear/base/bull	Fan chart; % and ₹/sq ft; confidence level (5-point) with explanation	P0 (rule-based range), P1 (model)
R-3	I see the drivers behind the forecast	List of catalysts + market trend + supply + risk, each with signed contribution %, probability and expected date; contributions sum to total forecast	P1
R-4	I can open the source of each catalyst	Each driver links to ≥ 1 source document (URL + archived copy + fetch date)	P0
R-5	I see risk flags	RERA complaints count, project delay vs RERA timeline, litigation (where sourced), flood/CRZ overlays	P1
R-6	I see comparable nearby projects	5 comparables with price and forecast	P1
R-7	I see historical accuracy for this city	Back-test metric displayed with methodology link	P1
R-8	Free users see a teaser; paid users see full detail	Free: today's range + outlook band + top 2 drivers. Paid: everything + PDF	P0

4.3 Catalyst explorer

ID	User story	Acceptance criteria	P
C-1	I can view all catalysts on a map, filterable by type and stage	Geometry (point/line/polygon) rendered; filter by type, stage, expected completion year	P0
C-2	I can open a catalyst detail page	Timeline of stage changes, budget, executing agency, sources, delay history, affected localities	P0
C-3	I can follow a catalyst	Alerts on stage change	P1

4.4 Reports, compare, portfolio, alerts

ID	User story	Acceptance criteria	P
P-1	I can buy and download a PDF report	Razorpay checkout (UPI/cards); PDF generated < 60 s (MVP: analyst-assisted within 24 h); GST invoice emailed	P0
P-2	I can subscribe to Pro	Monthly/annual plans; unlimited reports within fair use; cancel any time	P1
P-3	I can compare up to 4 properties	Side-by-side table + overlay chart	P1
P-4	I can add owned properties to a portfolio	Value + forecast tracking; monthly email summary	P2
P-5	I get alerts via email/WhatsApp	Opt-in per channel (DPDP consent); delivered within 24 h of the change	P1
P-6	Broker can white-label reports	Logo, contact details, client name; usage counted per seat	P1

4.5 B2B console & API

ID	User story	Acceptance criteria	P
B-1	Lender uploads a CSV of addresses and gets valuations + forecasts + risk flags	Up to 50k rows; match rate reported; results in < 1 h	P1 (pilot), P2 (GA)
B-2	Developer views an appreciation heatmap for a region	H3 grid heatmap, filters by horizon and segment, export	P2
B-3	Partner integrates via REST API	API keys, rate limits, usage metering, OpenAPI docs, 99.5% uptime	P2

4.6 Internal: data & analyst tools

ID	Requirement	Acceptance criteria	P
D-1	Crawlers fetch sources on schedule and archive raw files	Per-source schedule; hashing; diff detection; failure alerts	P0 (top sources), P1 (all)

ID	Requirement	Acceptance criteria	P
D-2	Document-AI extracts catalyst fields from PDFs/HTML	Fields: type, name, agency, geometry hint, budget, stage, dates; ≥ 85% field accuracy on a labelled set; confidence score per field	P1
D-3	Analyst review queue	Approve/edit/reject; side-by-side source view; audit log	P0
D-4	Entity resolution of projects/addresses	≥ 95% precision on a labelled set	P1
D-5	Manual override with reason	Overrides logged and shown in model governance reports	P1

5. Forecasting model requirements

ID	Requirement
M-1	**AVM**: hedonic + gradient boosting on registrations (primary) and listings (secondary, discounted for asking-price premium). Output P10/P50/P90. Target median absolute % error ≤ 10% in launch city
M-2	**Catalyst uplift model**: event-study / difference-in-differences on past projects (treatment = within X km of catalyst, control = matched areas). Uplift curve by catalyst type × distance band × city tier × lifecycle stage. Historical uplift is typically partly priced in at announcement; model the timing (announcement, construction start, opening) separately
M-3	**Completion probability**: per catalyst, by agency track record, stage, funding status, land acquisition %, litigation. Output a distribution of completion dates
M-4	**Market baseline**: city/segment trend from macro (rates, absorption, inventory months, income growth)
M-5	**Supply overhang**: penalty from RERA pipeline vs absorption rate in the micro-market
M-6	**Simulation**: Monte Carlo over completion dates and uplift uncertainty → fan chart
M-7	**Explainability**: additive decomposition of the forecast by driver
M-8	**Back-testing**: rolling-origin back-tests (e.g. forecast from 2016, 2018, 2020 and compare with realised prices); publish hit-rate within ±10% and MAPE
M-9	**Governance**: versioned models, reproducible training, drift monitoring, monthly re-scoring

6. Data model (core entities)

- `Property` (id, type, RERA id, developer, geometry, H3 cell, attributes)
- `Locality` / `MicroMarket` (polygon, city)
- `Transaction` (property/locality, date, value, area, ₹/sq ft, source)
- `Listing` (asking price, date, source)
- `Catalyst` (type, name, agency, geometry, budget, current stage, expected completion distribution)
- `CatalystStageEvent` (catalyst, stage, date, source document)

- `SourceDocument` (url, archived file, hash, fetched_at, license tag)
- `Forecast` (entity, model version, horizon, P10/P50/P90, drivers[])
- `User`, `Subscription`, `Report`, `Alert`, `Portfolio`, `Organisation`, `ApiKey`

7. Non-functional requirements

Area	Requirement
Performance	Report page TTFB < 1 s (precomputed forecasts); search < 300 ms
Availability	99.5% (B2C), 99.9% (API, Day 360)
Security	OWASP top-10, encryption at rest/in transit, SSO for B2B, role-based access
Privacy	DPDP Act 2023: consent, purpose limitation, data-principal rights, India data residency
Accessibility	WCAG 2.1 AA; English first, then Hindi and regional languages
SEO	Server-rendered pages per project/locality/catalyst; structured data
Observability	Logs, metrics, tracing; data-freshness dashboards per source

8. Legal & content requirements

- Every forecast shows: "Estimates are for information only and are not investment advice or a guarantee of future value."
- Never use "guaranteed returns" language; show ranges and confidence.
- Respect source terms; maintain a licensing register; attribute government sources.
- Risk flags must be factual and sourced (no subjective allegations about developers).
- Get legal review on whether any feature falls under SEBI investment-adviser rules or RERA agent-registration requirements (e.g. if brokerage is added later).

9. Analytics & experiment plan

- Funnel: search → report view → paywall → purchase → repeat.
- Experiments: free-tier depth, price points (₹499 vs ₹999), report-vs-subscription framing, WhatsApp alerts vs email.
- Track trust signals: source-link clicks, accuracy-page views, dispute submissions.

10. Release plan

Release	Contents
MVP (Day 1–30)	S-1, S-2, R-1 (locality), R-2 (rule-based), R-4, R-8, C-1, C-2, P-1 (analyst-assisted), D-1 (top sources), D-3
v1 (Day 90)	Building-level R-1/R-2 with models M-1, M-2, M-4, M-7, M-8; R-3, R-5, R-6, R-7; S-3; C-3; P-2, P-3, P-5, P-6; D-2, D-4, D-5; B-1 pilot
v2 (Day 360)	Multi-city; M-3, M-5, M-6, M-9; P-4; B-1 GA, B-2, B-3; plots/land

11. Open questions

1. Launch city: MMR (best data, most competition) vs Pune/Hyderabad (strong infrastructure pipeline, less noise)?
2. License transaction data from a vendor at the start, or build our own from IGR?
3. How deep should the free tier be to drive SEO and virality without killing conversion?
4. Should B2B (lenders) be prioritised earlier for revenue, at the cost of B2C brand building?
5. Which regional languages first?